From: Yardeni QuickTakes <yardeni-research@ghost.io>
Subject: The Bond Yield Spectrum
Date: 5 hours ago

The Bond Yield Spectrum

YARDENI QUICKTAKES

The Bond Yield Spectrum

By Ed Yardeni & Elias Griepentrog • 7 Oct 2026

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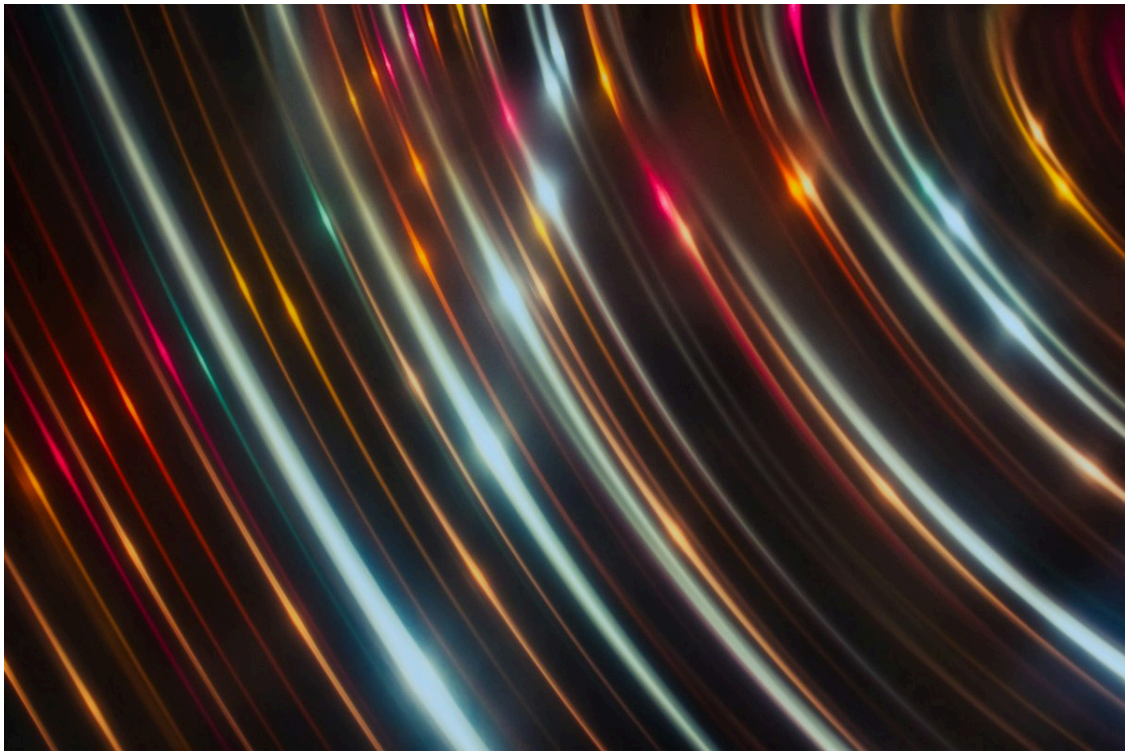
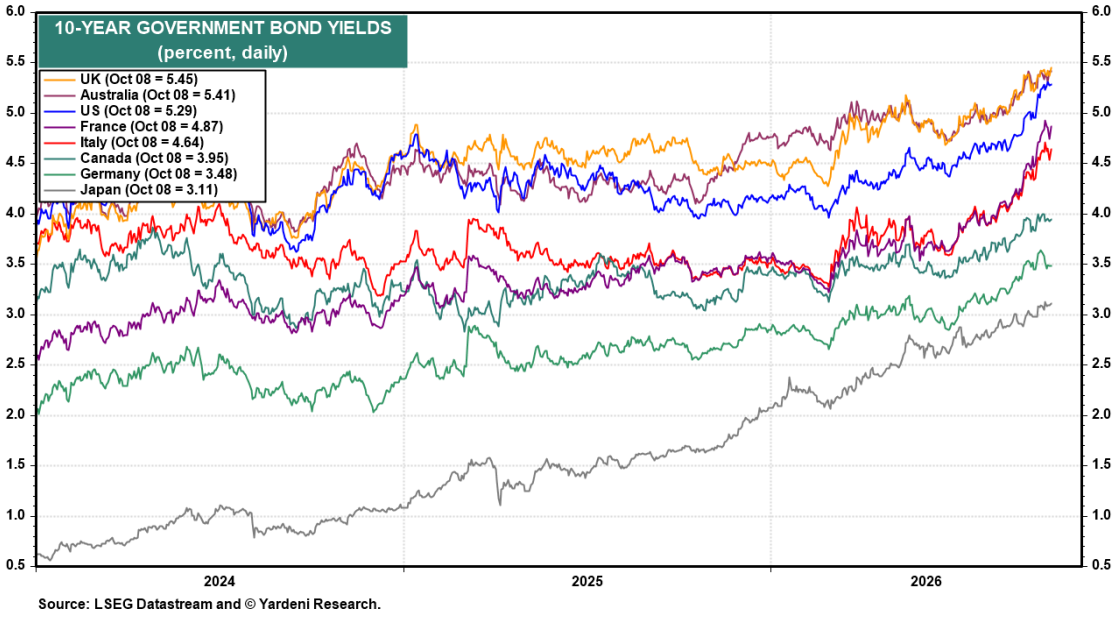


Photo by Adam Kring / Unsplash

I. From The Good To The Bad To The Ugly

The spectrum of explanations for the rise in bond yields worldwide includes relatively benign ones to downright dangerous ones (chart). On the good side of the spectrum, better-than-expected economic growth has driven US bond yields up to levels that are back to normal. On the ugly side, soaring bond yields in France are signaling a looming debt crisis. In between are bad scenarios that explain the synchronized rise in global

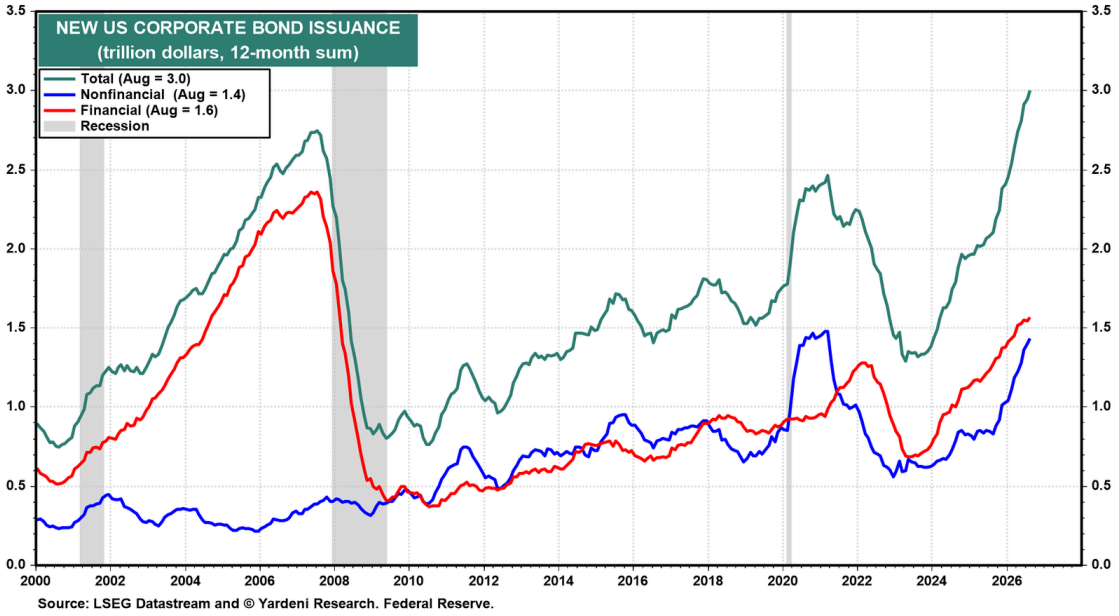
bond yields. These include the inflationary energy shock attributable to the war in the Middle East and the unwinding of the yen-carry trade.



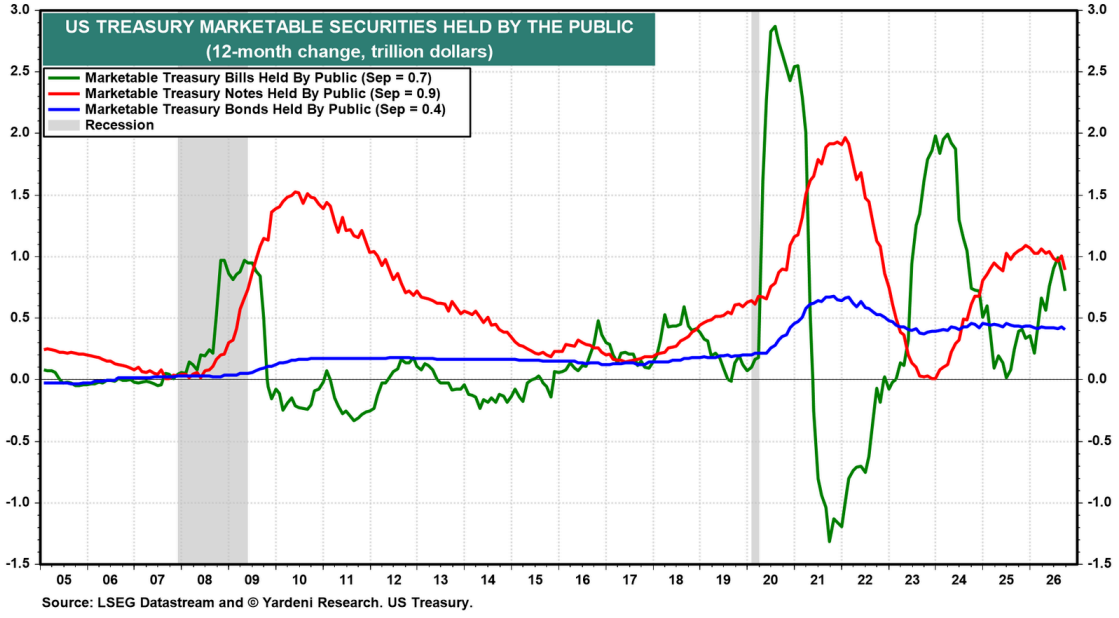
Let's have a closer look at the good, the bad, and the ugly:

(1) R-star rising. R-star, or the neutral interest rate, is the real interest rate consistent with low unemployment and price stability. It balances the supply of savings with demand for investment. Three forces are pushing it higher.

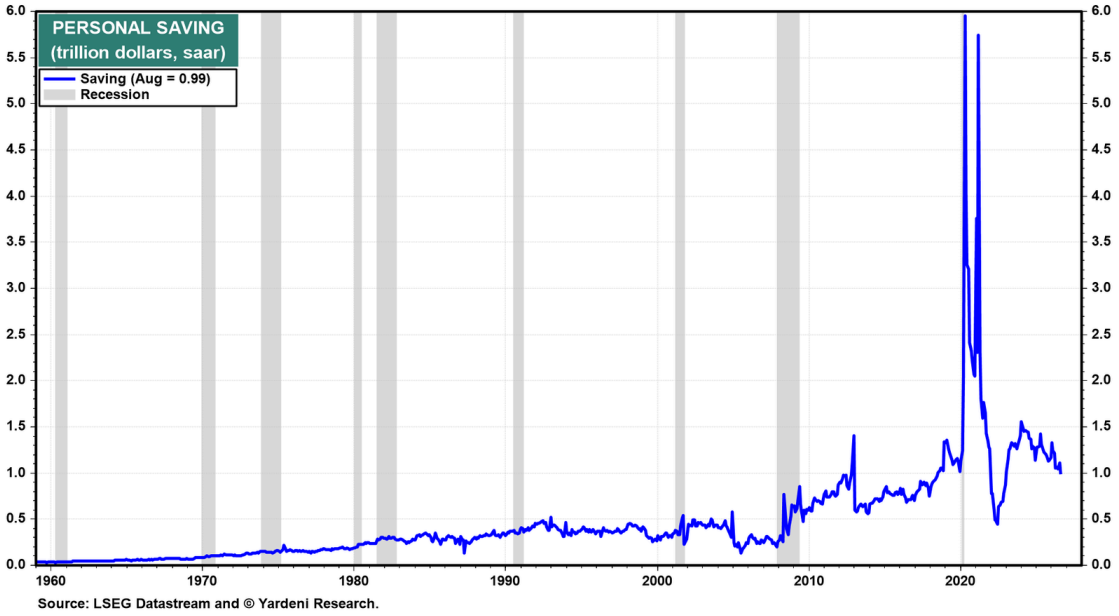
First, demand for capital is booming along with AI-related capital spending. Hyperscaler capex is expected to reach $750-$800 billion this year and $1.2 trillion next. Total US corporate bond issuance over the past 12 months through August was a record $3.0 trillion, including $1.4 trillion and $1.6 trillion issued by nonfinancial and financial corporations, respectively (chart).



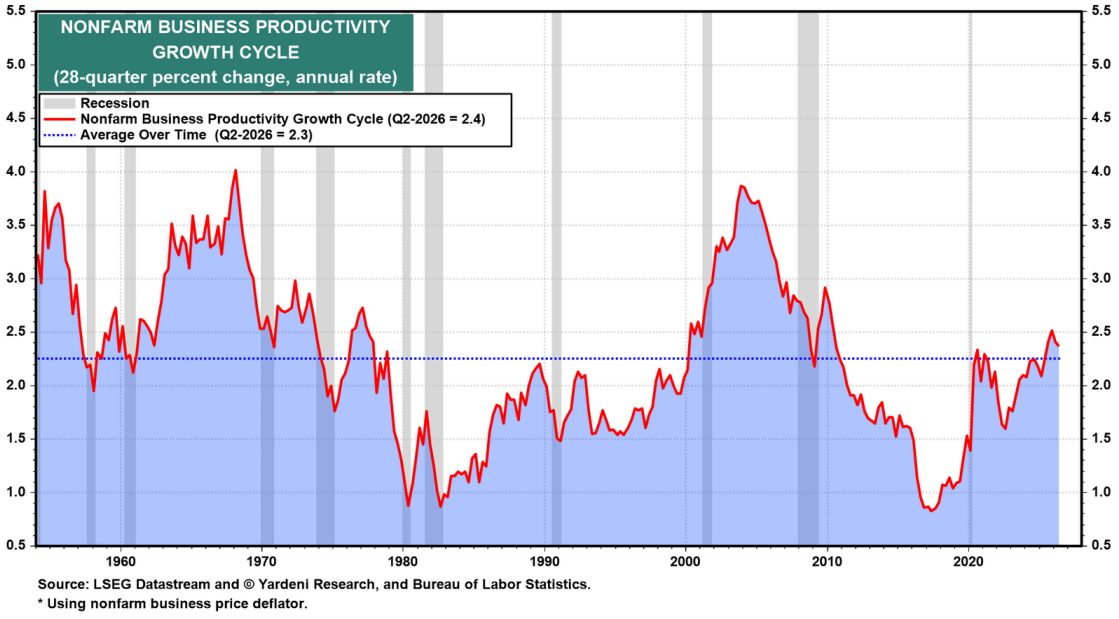
At the same time, US Treasury borrowing totaled $2.1 trillion over the past 12 months through September, including $1.3 trillion in notes and bonds (chart).



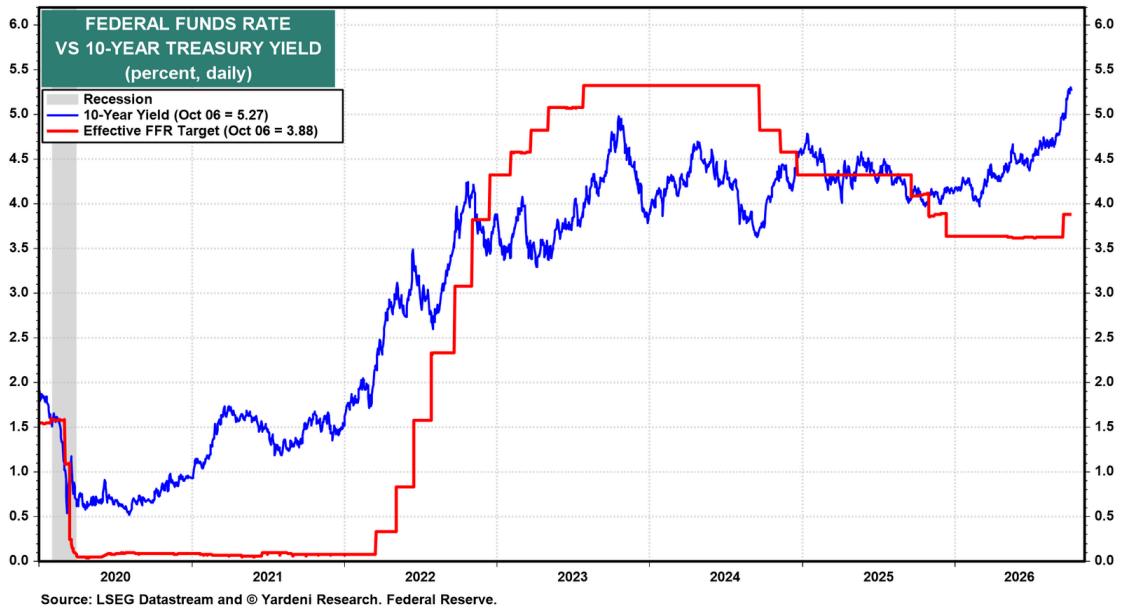
Second, national savings faces demographic headwinds as retiring Baby Boomers stop saving and draw down their net worth (chart).



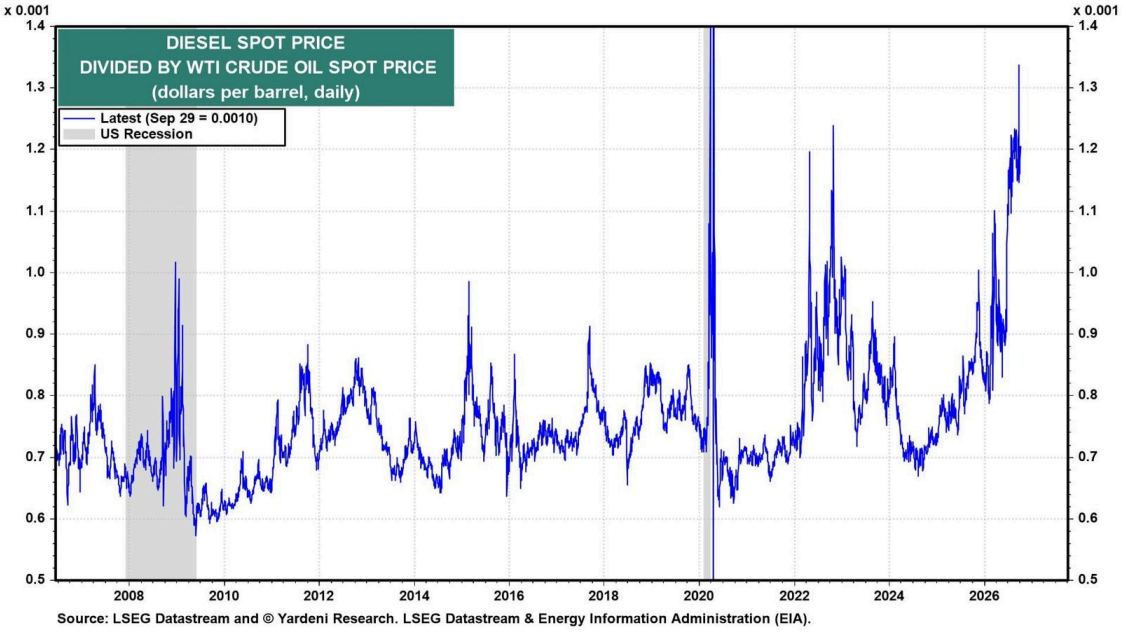
Third, stronger productivity growth may be raising the return on capital and the economy's sustainable growth rate (chart).



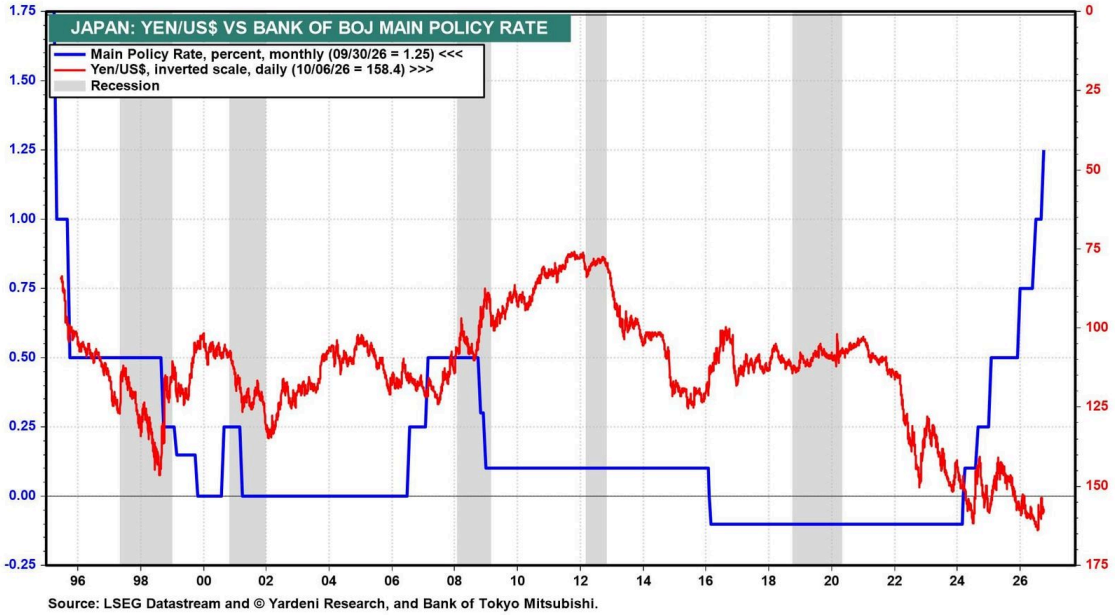
The Fed's 175bps cuts in the federal funds rate since September 2024 have been completely offset by a comparable increase in the 10-year Treasury bond yield since then. Even at the start of this year, most Fed officials believed that the FFR was modestly restrictive, i.e., that it exceeded the neutral rate. In September, they acknowledged their error by hiking the FFR by 25 basis points, as Fed Chair Kevin Warsh said they had removed a "dose of accommodation," implying that the FFR is actually below the neutral rate!



(2) Energy shock. Moving toward the middle of the spectrum are higher-for-longer oil prices. With US-Iran talks stalled and the US increasing its military assets, the risk of renewed escalation remains high. The longer energy costs stay elevated, the greater the risk they'll boost core inflation, forcing a more aggressive Fed response. If crude oil prices fall because more oil is getting through the Strait of Hormuz, diesel prices might remain elevated (chart).

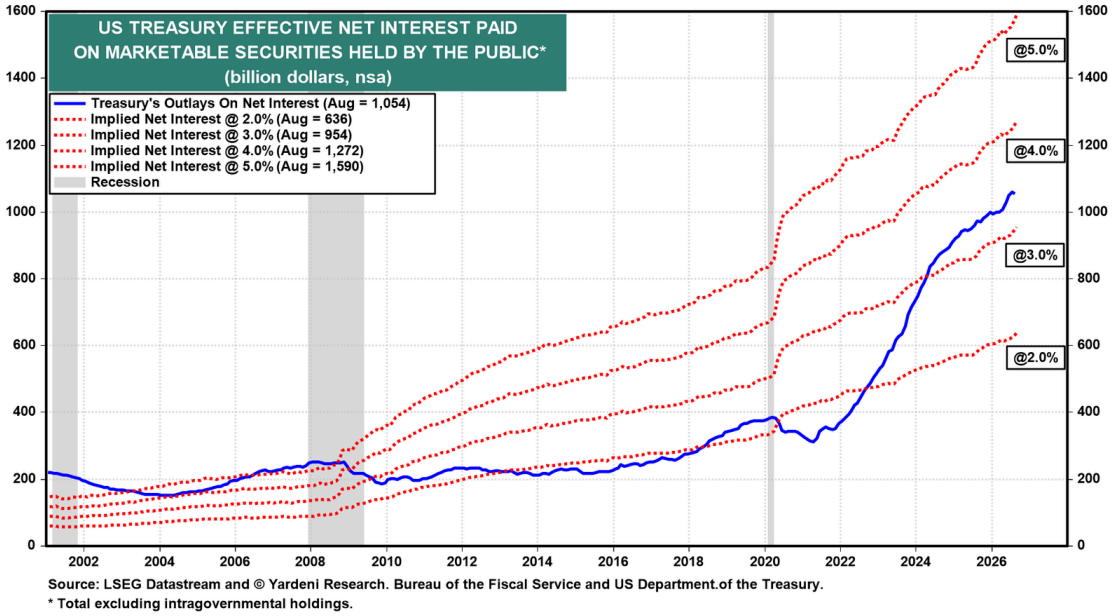


(3) Unwinding yen-carry trade. The yen-carry trade was based on ultra-low Japanese interest rates and a stable or weak yen. Now the BOJ is raising interest rates and the yen may be bottoming (charts). This is forcing traders to liquidate bond holdings previously financed with cheap Japanese money.

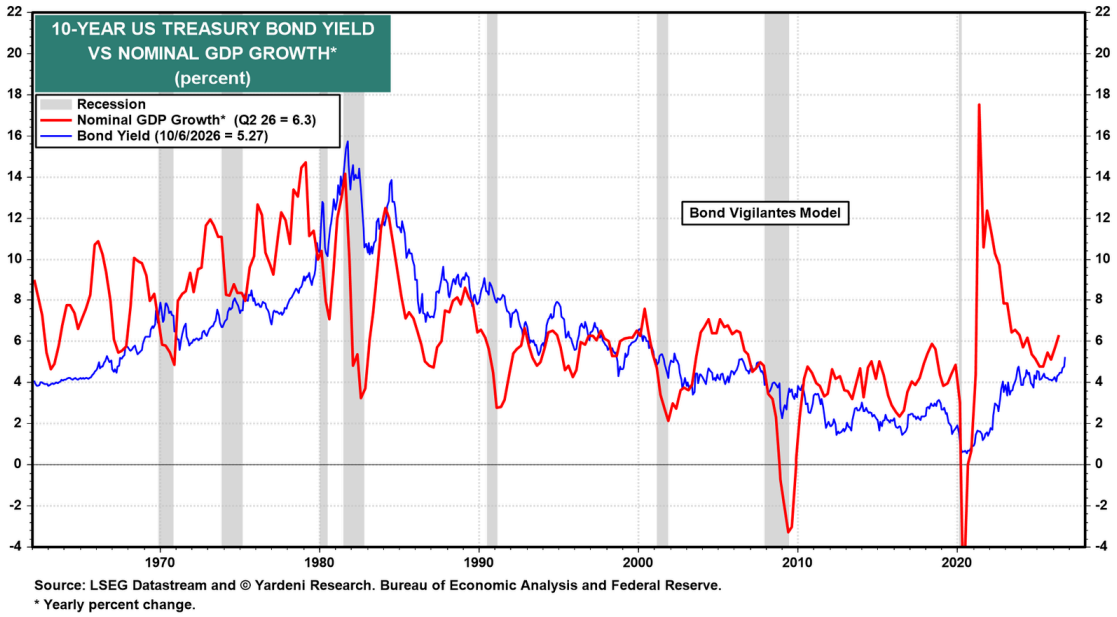


(4) Debt crisis. At the far end of the spectrum is the "Revenge of the Bond Vigilantes." Between the Great Financial Crisis and the Great Virus Crisis, major central banks' ultra-easy policies enabled governments to run large deficits. The Bond Vigilantes were powerless. But now they are more powerful than ever because government debt is at record highs. Central banks have been forced to raise their policy rates as inflation has been more troublesome in recent years. Bond yields have soared this year worldwide as inflation rebounded and fiscal-risk premiums have risen.

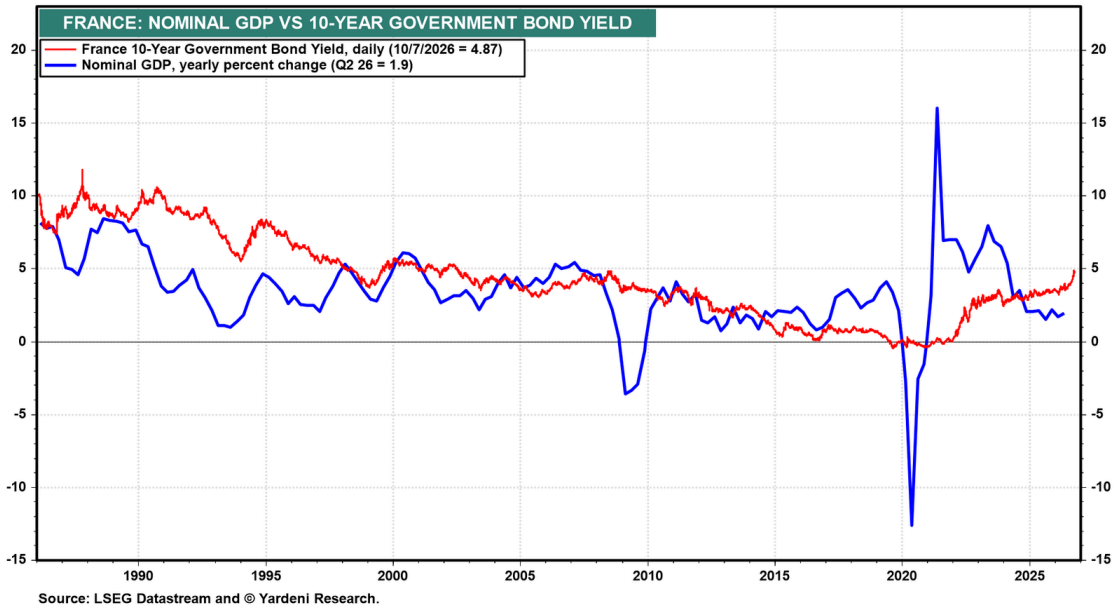
The danger is a self-reinforcing debt spiral: Higher yields raise interest costs, widen deficits, require more borrowing, and push yields higher still (chart). And, of course, higher yields increase the risk of causing a recession, which would exacerbate any debt crisis.



In our view, the US should remain in the higher R-star end of the spectrum. We would start to worry about the debt-crisis danger zone if the US bond yield rises above nominal GDP growth (chart).



In France, on the other hand, the Bond Vigilantes are pushing the 10-year government bond yield well above nominal GDP growth (chart). The French could be toast if yields continue to soar.



II. The Fed Minutes

The latest Fed minutes paint a familiar picture: a resilient economy, a firm labor market, and inflation that remains too high. We drew three key takeaways from the minutes of the Fed's September 15-16 meeting.

(1) Restrictiveness. Several participants said the current policy rate was not restrictive or only mildly restrictive, while a couple explicitly raised their estimates of the neutral federal funds rate. That fits neatly with our R-star story.

(2) Macro backdrop. The Fed saw a resilient economy: real GDP is expanding at a solid pace, consumer spending has firmed, AI investment remains robust, and the labor market is close to maximum employment. At the same time, inflation remains elevated and risks were still skewed to the upside.

(3) Rate outlook. All participants backed September's 25bps hike, and most judged that another increase would likely be appropriate by year-end.

💡

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Subject: Following Nebius, AWS is hiking EC2 capacity blocks listed price by 15%
Date: 2 hours ago

Following Nebius, AWS is hiking EC2 capacity blocks listed price by 15%

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Following Nebius, AWS is hiking EC2 capacity blocks listed price by 15%

AWS 調漲 EC2 預留算力價格 15%：鎖定 80 億美元輝達晶片交易，公有雲定價權全面重構

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Executive summary

即使最近我們觀察到部分AI伺服器產業鏈下游如ODM組裝廣達，滑軌川湖，散熱系統雙鴻，及均熱片健策，9月份訂單有明顯調整，3Q26營收小幅不如市場預期，可能因為短期電力短缺及短料缺貨卡長料庫存調整，但AI數據算力中心基礎建設的價格調整似乎還未停歇，先是有 Nebius 於10/1日起全面調漲 AI GPU 租用價格達 17-21%，CPU 達25%，Memory 達41%（請參考之前報告：17-21% On-demand AI GPU pricing hike to shift value from LLM companies to cloud vendors），現在又有 AWS 於10/7日起調漲 EC2 容量區塊牌價達15%，但使用客戶大語言模型公司 OpenAI, Anthropic 卻持續降模型每百萬 Tokens 價格衝市佔，***我們重申在電力短缺及短料缺貨卡長料庫存造成既有算力稀缺的同時，短期市場價值將從賣 tokens 的大語言/專家語言模型公司轉到 AI 數據中心算力基礎建設公司***。



What’s new?

過去 24 小時內，亞馬遜雲端運算服務（AWS）官方更新了其 EC2 算力目錄，正式自 2026 年 10 月 7 日起上調指定高階 GPU 預留容量區塊（EC2 Capacity Blocks）牌價，實質調漲幅度高達約 15%。這項定價調整與亞馬遜正在推進的 80 億美元輝達 GPU 專項結構化融資案同步浮出水面。在此之前，專用雲龍頭 CoreWeave 與 Nebius 甫於 10 月 1 日將 H100 及 B200 現貨時租價格調漲 17% 至 21%。AWS 此次調價確認了公有雲龍頭不再打價格戰，而是全面跟進調價，彰顯出在企業長週期 Agent 與前線訓練負載湧入下，頂級算力供給依舊供不應求。

付費訂閱戶可分享誰有利，誰受影響，Outside the box 想法，建議，什麼是EC2 capacity blocks

Subscribed

Positively affect what supply chain vendors and why?

- **超大規模公有雲巨頭：**亞馬遜（Amazon AWS）、微軟（Microsoft Azure）、Alphabet（Google Cloud）。
- **Neocloud 外溢雲運營商：**CoreWeave（CRWV）、Nebius（NBIS），Oracle, SpaceX/SI。
- **企業級 GPU 雲端資源優化軟體：**Cast AI、Run:ai。
- **原因分析：**預留算力價格上調 15% 將直接提振 AWS 雲端運算業務的平均單元營收（ARPU）與營運利潤率，消除了市場對雲巨頭巨額 Capex 侵蝕自由現金流的擔憂；同時專用雲的提價策略獲得 Hyperscaler 背書，支撐了整體行業的毛利擴張，我們相信其他公司也會陸續跟進。

Negatively affect what supply chain vendors and why?

- **預算受限的中小型 AI 開發者與微調新創。**
- **尚未鎖定長期預留合約的二線軟體公司。**
- **原因分析：**算力租金成本上揚 15% 將直接壓縮非頭部 AI 應用公司的現金跑道（Runway），迫使缺乏強勁變現能力的新創加速整合或退出市場。

Our outside the box non-consensus thinking

市場直覺認為雲端巨頭提價可能引發客戶流失或轉向自建資料中心。然而我們的非共識洞察是 AWS 此次提價並非單純的成本轉嫁，而是算力期貨化與優質客戶篩選機制優化。在 EC2 Capacity Blocks 機制下，客戶鎖定的是「未來特定時段確保可用的確定性算力」。在全美資料中心正面臨超過 100 項地方性建置禁令（Moratoriums）與電網審批塞車的背景下，「即時可用的電力與算力」成為絕對稀缺品。AWS 藉由調價 15%，有效過濾了投機性短期負載，將關鍵算力優先供給具備長期訂閱付費能力的高價值企業客戶（如生成式醫療、高頻金融與國防客戶），從而鎖定長達 3 至 5 年的現金流可預測性。

Our advises and risks analysis

- **超配雲端巨頭母公司股權：**AWS 提價確認了強大的定價權，亞馬遜（AMZN）預計將在2027-2028展現高於預期的利潤率修復。
- **持有 GPU 虛擬化與排程優化軟體股：**隨著時租成本攀升，企業將加大採購動態排程與工作負載優化軟體，推升相關供應商營收。
- **客戶架構優化導致 Token 效率提升：**推論優化演算法可能減少總 GPU 租賃時長，這對雲巨頭是漲價後可能發生的風險。
- **總體經濟逆風壓抑 IT 支出：**若企業削減 2027 年 IT 資本預算，可能影響高價預留容量的續約率。

主要新聞來源鏈接（過去24小時國際要聞）：

• INDmoney: Amazon’s $8B Nvidia Chip Deal: AI Infrastructure Financing & Price Hikes
• 24/7 Wall St.: Amazon Is Spending Another $1 Billion on AI Data Centers
• Value Add VC: Big Tech AI Capex Hits $730B in 2026, Amazon Raises to $220B

附註：EC2 及EC2 capacity blocks 是什麼?

Amazon EC2（Elastic Compute Cloud）是亞馬遜雲端運算服務（AWS）最核心的 IaaS（基礎設施即服務）產品，簡單說就是「在雲端租用的虛擬伺服器」。企業與開發者不再需要自行採購、架設或維護實體機房與伺服器硬

體，只要透過網路，幾分鐘內就能按需開啟或關閉任意運算規格的主機，並依實際使用時間計費。

一、 EC2 的核心概念與結構

1. **實例（Instances）**：
    - EC2 中的虛擬機器統稱為「實例」。
    - AWS 依據不同的工作負載，將實例分成不同家族，例如：
        - **通用型（M 系列、T 系列）**：平衡 CPU、記憶體與網路，常用於一般網站、後端 API。
        - **運算優化型（C 系列）**：高單核效能，適用於批次處理、分散式分析。
        - **記憶體優化型（R 系列、X 系列）**：配備海量 RAM，用於記憶體資料庫（Redis/SAP HANA）。
        - **加速運算型（P 系列、G 系列、Trn 系列）**：搭載 NVIDIA 高階 GPU（如 H100、B200、B300 等）或 AWS 自研晶片（Trainium/Inferentia），專門用於 AI 大模型訓練與推論。
2. **彈性（Elastic）**：
    - 資源可以**水平擴展（Auto Scaling）或垂直升降級**。例如電商遇到購物節流量爆發，系統可自動從 5 台擴增至 50 台，活動結束後自動銷毀，避免資源閒置浪費。

二、 常見的計費與購買模式

傳統上，EC2 的採購模式主要分為：

- **隨需（On-Demand）**：隨開隨用、隨關隨停，按秒/小時收費，價格最高但彈性最大。
- **節約計畫 / 預留（Savings Plans / Reserved Instances）**：承諾使用 1 年或 3 年，換取高達 40%–70% 的折扣。
- **競價實例（Spot Instances）**：利用 AWS 當下閒置的閒置運算容量，折扣極高，但隨時可能被 AWS 回收中斷。

三、 什麼是「EC2 Capacity Blocks」？

在 AI 浪潮爆發後，高階 GPU 供不應求。許多 AI 新創或研究團隊買不起 1~3 年的長約（Reserved），但用隨需（On-Demand）又常因機房沒卡而啟動失敗，用 Spot 則可能在模型訓練到一半時被踢掉。

因此 AWS 推出了 **EC2 Capacity Blocks for ML（機器學習容量區塊）**：

- **運作機制**：允許客戶提前預約未來某個特定日期、特定時段（最長可預約數週前，租期由 1 天至數十天不等）的頂級 GPU 算力叢集（如 P5、P6

等高階實例)。

- **保證取得**：在預約的時間區間內，AWS 保證提供 100% 確定可用的 GPU 叢集與高速低延遲網路（EFA)。
- **調價意涵**：AWS 上調 Capacity Blocks 牌價（特別是針對 B200/B300 等次世代 Blackwell 架構機型)，反映出市場對超高階 AI 訓練與推論算力的強烈需求與稀缺性，雲端巨頭也藉此反映其龐大的硬體採購、先進封裝及水冷機房建置成本。

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WEEKLY WEBCAST: Bond Yields On The Spectrum: From Good To Ugly (with special guest Nick Raich)

By Ed Yardeni • 7 Oct 2026

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Is the bond selloff the Revenge of the Bond Vigilantes, i.e., a looming debt crisis? Not yet. Despite the recent surge, the 10-year Treasury yield remains below the growth rate of nominal GDP. That points toward an upward repricing of the neutral interest rate. Ed and Elias examine the spectrum of forces behind higher yields, from rising R* to a potential debt-crisis scenario. The real warning sign would be if the yield rises above nominal GDP growth. In addition, they consider two other spectrum scenarios: the war in the Middle East and the yen-carry trade. … Also: The latest labor market data suggest resilient but noninflationary strength.

🔒

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Subject: Elon Musk denies TSMC to build and run TeraFab for SpaceX/Tesla
Date: 16 hours ago

Elon Musk denies TSMC to build and run TeraFab for SpaceX/Tesla

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Elon Musk denies TSMC to build and run TeraFab for SpaceX/Tesla

路透社說馬斯克否定台積電將為 SpaceX/Tesla 建廠及經營 TeraFab，頂多是分租場地

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What's new?

Elon Musk 表示 Tesla 與 SpaceX 將自行興建、營運 Texas TeraFab，並強調 TSMC 最多可能分租其中一部分；Intel CEO Lip-Bu Tan 表示 Intel 作為夥伴仍將參與。這是營運主體的明確澄清，但並非晶圓廠技術或投資決策的完整公告。

主要新聞： Reuters，2026-10-07 13:13 台北

Positively affect what supply chain vendors and why?

若專案落地，美國 fab EPC、無塵室、電力／水處理、Applied Materials、Lam Research、KLA、ASML、EDA／IP 與先進封裝供應商具選擇權。Intel 若提供製程技術、營運人才或代工支援，可提升產能利用率與先進製程外部驗證。

Negatively affect what supply chain vendors and why?

「TSMC 出資、擁有、營運」的題材被直接否定，現在 Tesla／SpaceX 必須全面承擔良率、人才、污染控制、供應商認證與資本密度。若 TeraFab 只做成熟製程或後段整合，其對先進邏輯設備與材料的拉動會遠低於市場期待。

Our outside-the-box, non-consensus thinking

台積電分租 TeraFab 其中一部分是很奇怪的決定，依照過去經驗，台積電在先進製程應該會自己擴產，營運，頂多拿些 SpaceX/Tesla 資金，實在沒有理

由去分租 TeraFab 一部分廠房。除非TeraFab 只想先做晶圓製造，但把 CoWoS/3D IC 先進封裝部分先交給台積電負責，但這對台積電而已除了可能拉低毛利率，也培養客戶在晶圓製造上產生合理現金流再加速追趕技術，實在不合理。

Our advice to investors

在看到六項硬證據前，只給選擇權價值：法人實體與股權、總 capex、製程節點、wafer starts per month、首批設備 PO、量產／良率責任。TSMC 分租目前不應估入其長期營收；Intel 參與也不等於已取得製造訂單。

Risks to our analysis

Musk 的專案範圍與合作模式仍可能有改變；TSMC、Intel 均未立即回應。美國補貼、出口管制、設備交期、人才、電力與水資源都可能改變經濟性；若 TeraFab 僅是談判工具，供應鏈訂單可能不會發生。

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2028 Should Be A Good Year



YARDENI QUICKTAKES

2028 Should Be A Good Year

By Ed Yardeni & Elias Griepentrog • 6 Oct 2026

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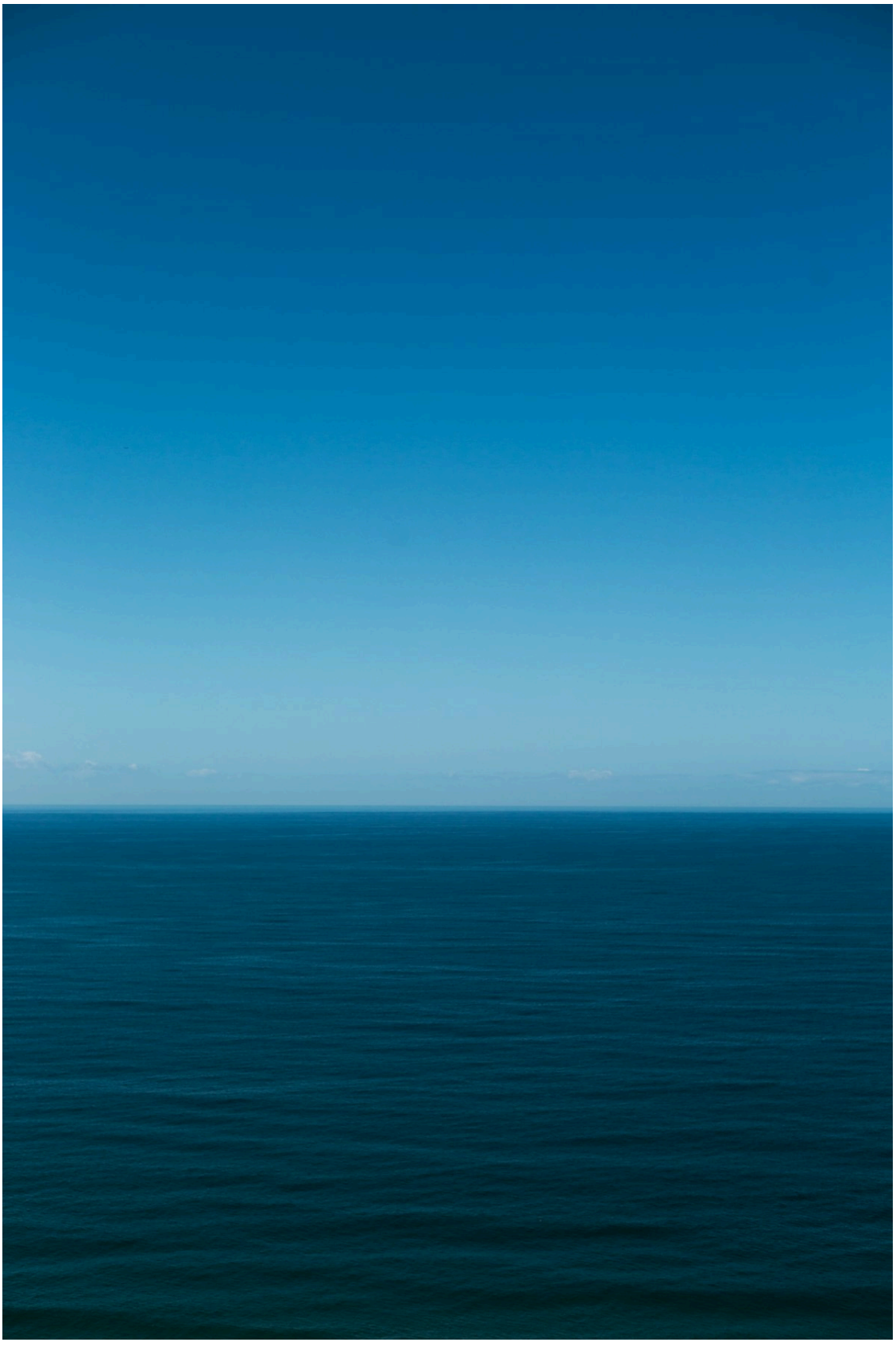


Photo by Patrick Fore / Unsplash

I. Blue skies ahead for earnings?

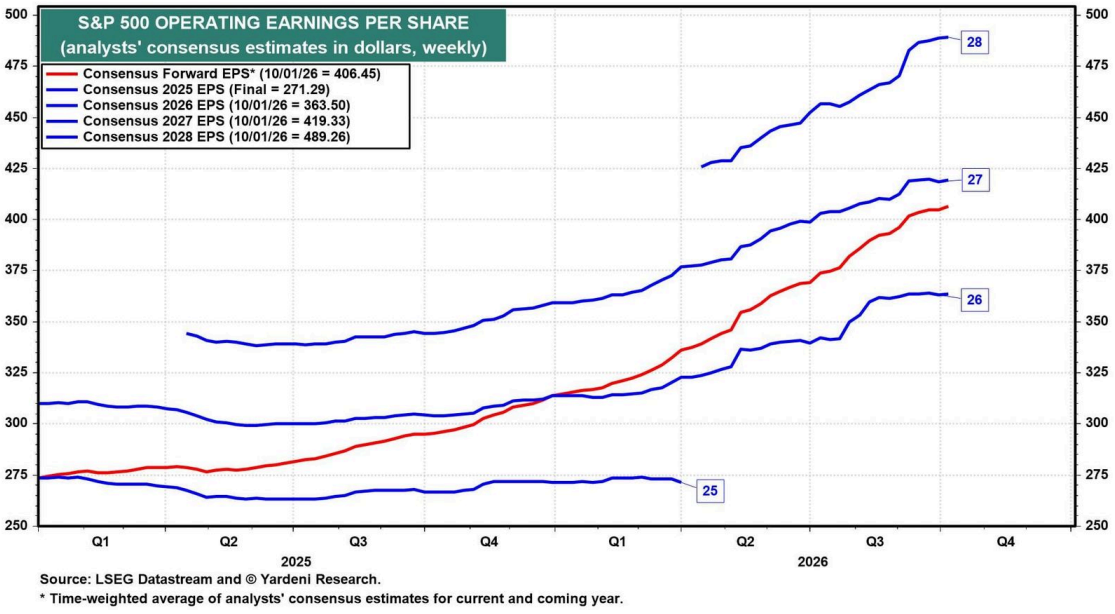
We are now seven years into our Roaring 2020s thesis, which remains our base case for the rest of the decade, and it has played out very well so far. When we first introduced the idea in August 2020, the prospect of another Roaring '20s sounded a bit delusional. We argued that, as in the

1920s, rapidly proliferating technologies would drive a productivity boom that would support stronger economic growth while containing inflation, boosting real wages, widening profit margins, and lifting corporate earnings.

Since the beginning of the current decade, the economy's resilience has repeatedly defied the naysayers, as productivity growth has increased and corporate earnings have continued to climb. With only three full years left in the decade, one of the clearest manifestations of our Roaring 2020s thesis has been what we call FEMO, or Fabulous Earnings Momentum.

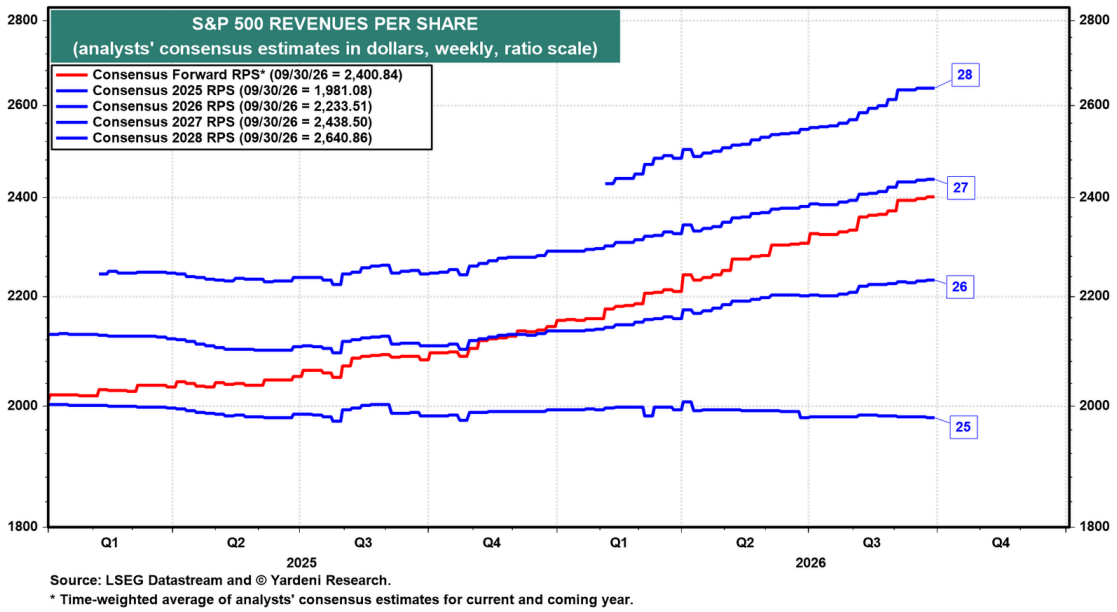
FEMO is clearly visible in industry analysts' earnings expectations. They currently expect S&P 500 operating earnings per share to rise to about $364 per share in 2026 and $419 in 2027, implying growth of roughly 15% in both years (chart). Toward the end of Q3 each year, we add another year to our Earnings Squiggles framework. This year’s addition is 2028, with earnings projected to rise another 17% that year to about $489 per share.

For now, the analysts are more bullish than we are about earnings prospects over the remainder of the decade. We are projecting $450 per share in 2028, $475 in 2029, and $500 in 2030. If they are right, our S&P 500 target of 10,000 will be achieved well before the end of 2029!

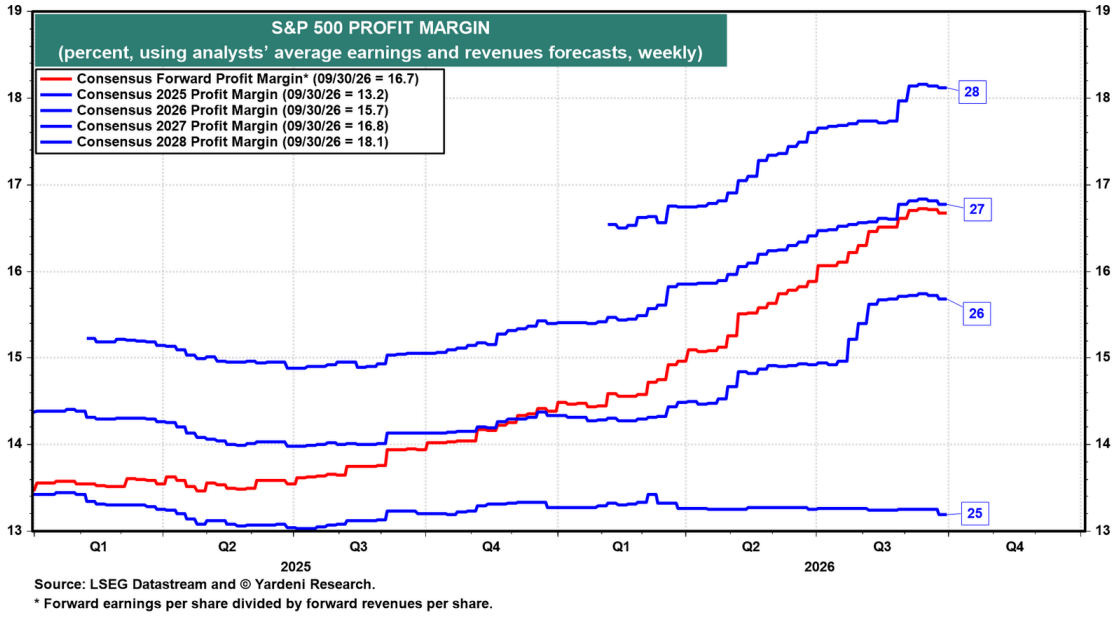


The extended earnings boom is driven by strong revenue growth expectations. Industry analysts expect S&P 500 revenues per share to

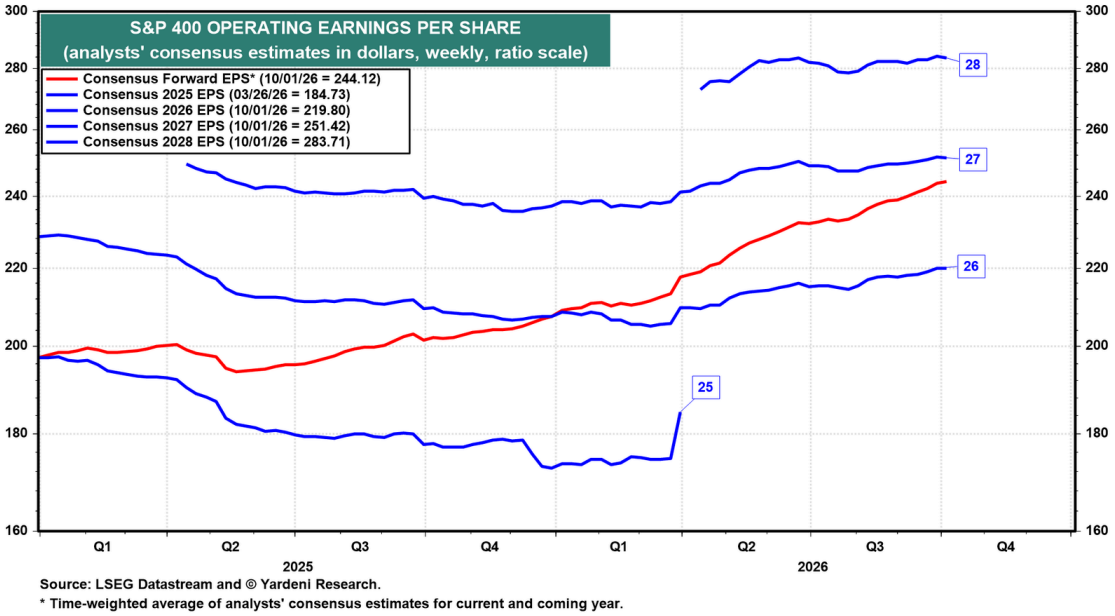
rise 12.7% in 2026, 9.2% in 2027, and another 8.3% in 2028 (chart). Those expectations seem too exuberant to us.



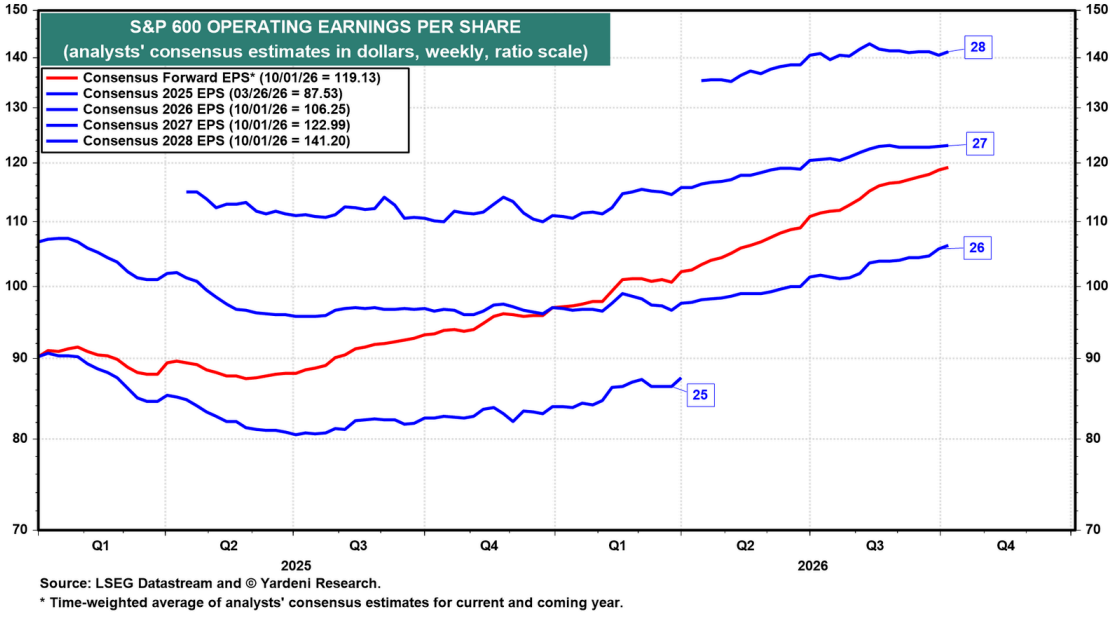
Profit margins provide the other driver of the earnings story. Analysts expect the S&P 500 profit margin to rise from 13.2% in 2025 to 15.7% this year, 16.8% in 2027, and a record 18.1% in 2028 (chart). That is what we would expect in our Roaring 2020s scenario, with stronger productivity growth widening profit margins.



The earnings outlook is strong beyond the S&P 500 as well. Industry analysts expect S&P 400 MidCap operating earnings per share to rise roughly 19% in 2026, 14% in 2027, and 13% in 2028 (chart).

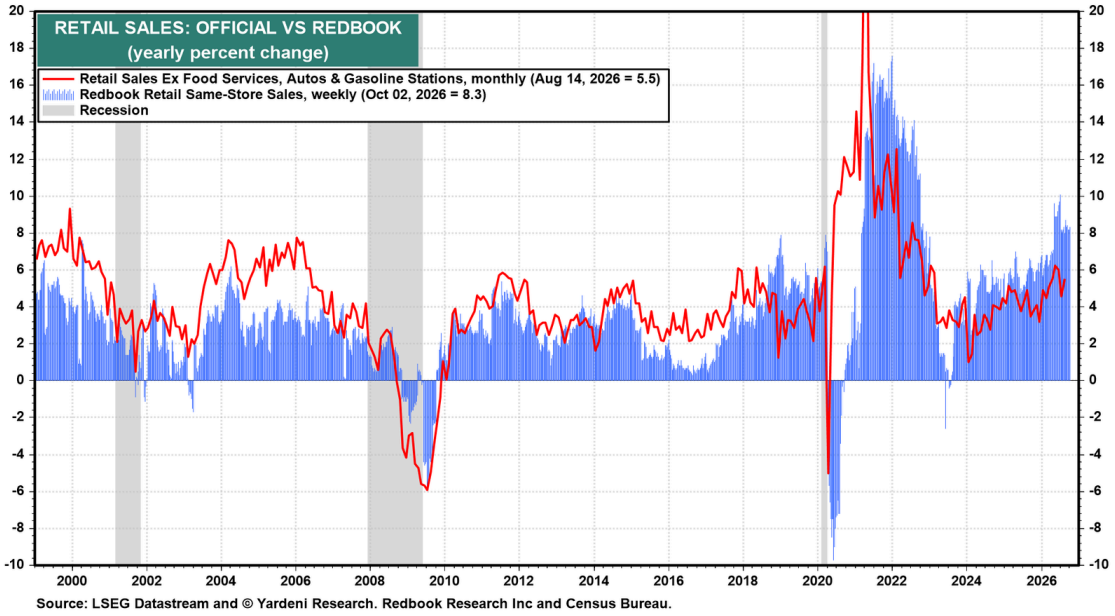


S&P 600 SmallCap earnings are expected to increase about 21% this year, 16% in 2027, and 15% in 2028 (chart). That suggests FEMO is broad-based.

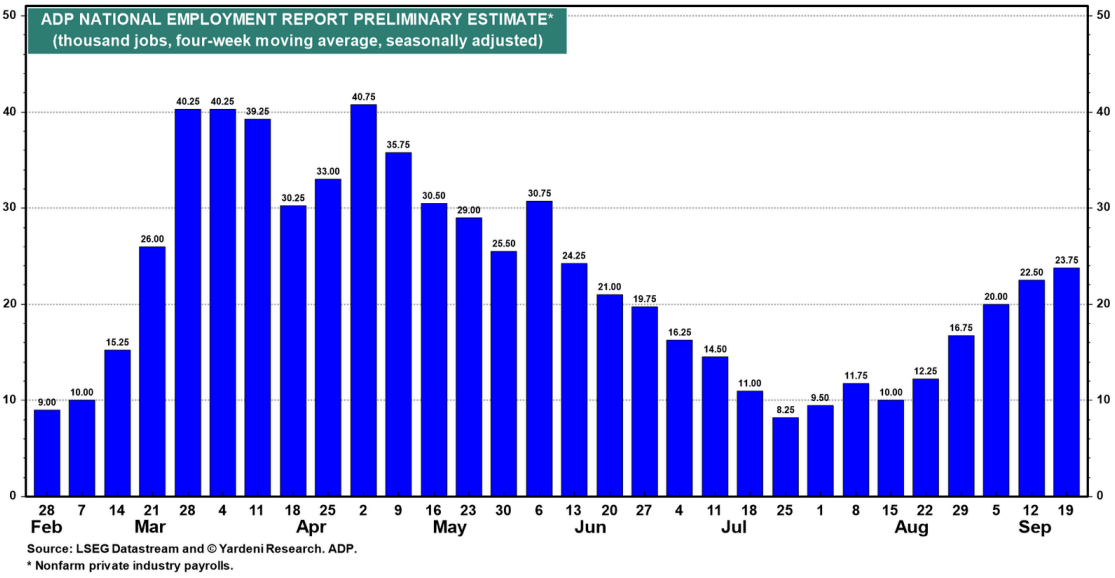


II. Blue skies ahead for the economy?

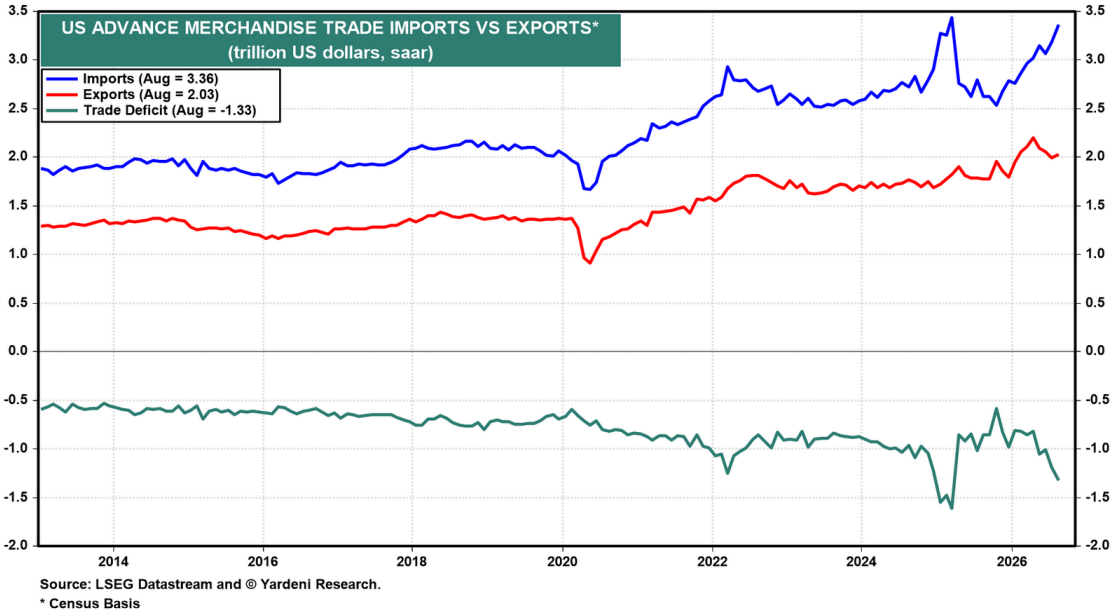
(1) Consumer spending. The latest Redbook data provide yet another sign of continued strength in consumer spending. Redbook same-store sales rose 8.3% y/y during the week of October 2, well above the 5.8% average for 2025 and underscoring the consumer's continued resilience (chart).



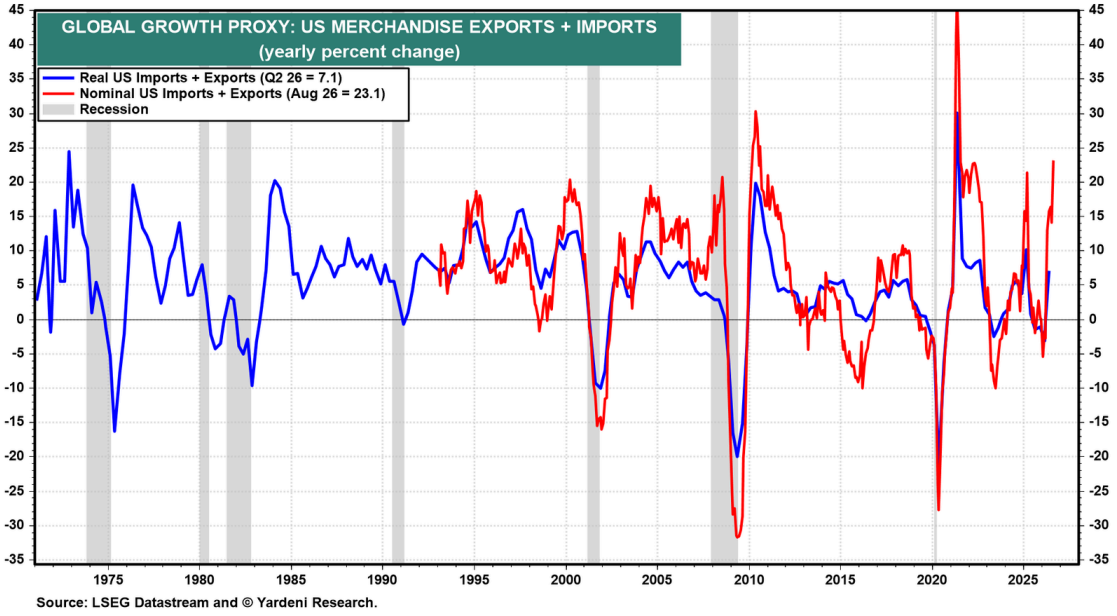
(2) Labor market. ADP estimates that private-sector payrolls rose 23,750 in the week through September 19, up from an upwardly revised 22,500 the prior week and marking the fifth consecutive acceleration in weekly job gains (chart). It was also the strongest weekly increase in three months. The ADP numbers for private industry payrolls make more sense to us than the Bureau of Labor Statistics' payroll employment data.



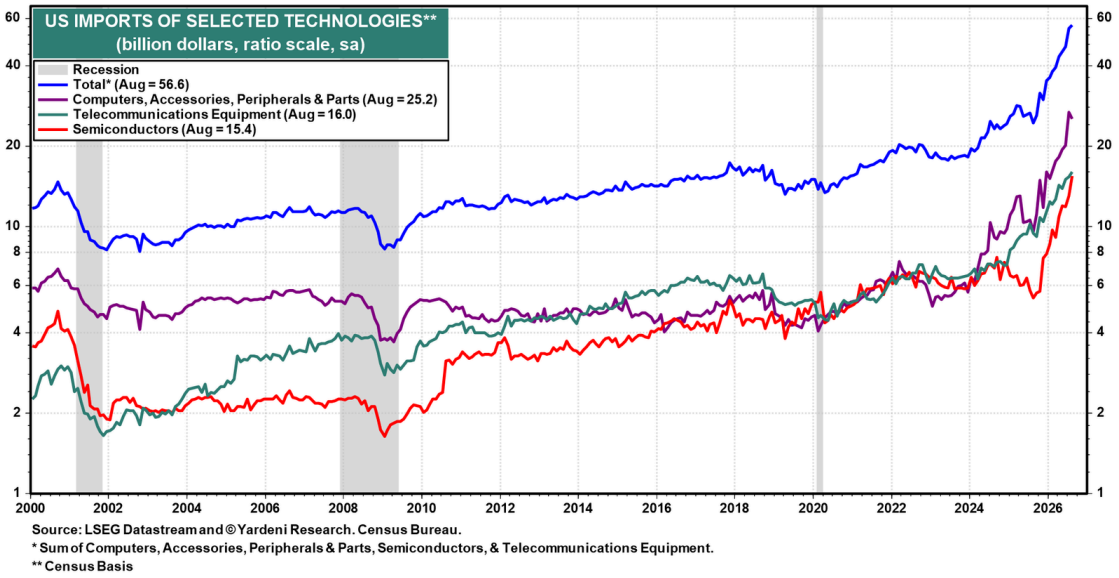
(3) Trade balance. The latest US trade data point to still strong US and global economic growth. US imports rose 4.3% m/m and 23.6% y/y in August, accelerating from July, while US exports increased 1.4% m/m and 12.3% y/y. Imports reached a $3.36 trillion annualized rate, versus $2.03 trillion for exports, leaving a merchandise trade deficit of $1.33 trillion annualized (chart). The gap reflects exceptionally strong US demand for imported goods.



Nominal and real US merchandise exports plus imports rose 23.1% y/y and 7.1%, respectively, in August (chart). That suggests global trade remains firmly in expansion territory and is consistent with strong global economic growth.



Selected technology imports reached a record $56.6 billion in August (chart). The AI buildout is highly import-intensive and requires large volumes of semiconductors, servers, networking equipment, and other hardware produced overseas.



💡

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Date: 9 hours ago

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Shifted the focus back to energy power vendors again

在谷歌與Constellation 簽署兩份長期合約後，市場明顯將注意力轉回電力能源產業

ANDREW LU ON GLOBAL SEMI/TECHS
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What's New?

Google 與美國最大核電業者 Constellation Energy 達成指標性長期採購協議，簽署 20 年期 43 億美元核能大單，總供電承諾達 3,590 MW。

雙軌合約結構（增容 PPA + 長期基載電力）：

890 MW 功率增容 PPA：為期 20 年，Google 直接資助 Constellation 位於伊利諾州、賓州與紐澤西州的 11 座現有核電機組進行翻新與技術升級（更換蒸汽發生器、渦輪機與控制系統），相當於憑空增添一座大型反應爐的發電容量，預計 2028 年起陸續供電至 2033 年全面上線。

2,700 MW 電網電力供應協議：為期 15 年，鎖定 PJM 電網內的穩定電力配額（請參考附註了解PJM電網架構），為 Constellation 機組提供長期現金流保障。

開創「自帶電力（Bring Your Own Power, BYOP）」防禦機制：針對資料中心擴建引發「科技巨頭推高民生電費」的批評，本案採用 PJM 電網架構，由 Google 自掏腰包承擔核電站設備升級的巨額資本支出，主打不將成本轉嫁給一般居民用電戶。

科技巨頭核能策略可能轉向「既有設施增容與重啟」：繼微軟重啟三哩島（Three Mile Island）、Meta 採購 Clinton 核電廠電力、亞馬遜合作 Talen

Energy 後，Google 也從長遠的 SMR（小型模組化反應爐）進一步落實到即時可行的「既有核電廠增容」與「退役電廠重啟」路線。

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What companies can be benefited and why?

核電與獨立發電業者（Independent Power Producers, IPP）：

Constellation Energy (CEG)：最直接受益者，獲得確定性極高的 43 億美元投資承諾與 20 年長期現金流，大幅提升估值倍數。（請參考之前深度報告：What power energy names are benefited more from 100GW data center expansion? 8 names in comparison ）

Vistra (VST)、Talen Energy (TLNE)、PSEG：全美非受監管核電與基載電力資產被全面重估，從過去低成長的公用事業轉型為 AI 時代具戰略溢價的稀缺基載資產。

重型發電與核電設備供應商（Power & Industrial OEMs）：

GE Vernova (GEV)、Westinghouse（Cameco / Brookfield 持股）、西門子能源（Siemens Energy）：核電增容（Uprates）的核心在於更換汽輪機、大型變壓器、蒸氣發生器與數位監控系統，直接帶動重型電力設備供應鏈龐大訂單。

核燃料供應鏈（Uranium Miners & Enrichment）：

Cameco (CCJ)、Centrus Energy (LEU)、Uranium Energy (UEC)：機組增容運轉及電廠延役延長了燃料消耗週期，讓天然鈾（U3O8）與低濃縮鈾（LEU）的長期結構性需求更加穩固。

超大規模雲端巨頭（Google / Alphabet）：

在 PJM 這個全美資料中心最密集且電網負載最吃緊的區域，提前鎖定長達 20 年的無碳基載電力，避免因電網壅塞而被迫延後 AI 伺服器機房上線，同時捍衛其 2030 年 24/7 零碳排（CFE）的承諾。

What companies can be negatively affected and why?

缺乏儲能的純綠能業者（Pure-play Wind / Solar Developers）：

AI 資料中心要求 99.999% 的 24/7 恆定供電。純風光發電的間歇性在科技巨頭預算配置中逐漸失寵，科技公司的購電優先順序正轉向核能與地熱，傳統未配儲能的純光伏/風電開發商的議價空間遭到壓縮，除非再配合儲能系統一起打包出售。

二線雲端業者與中小規模資料中心業者（Tier-2 CSPs & Mid-size Colocations）：

電力正取代晶片成為算力擴張的最大瓶頸。巨頭憑藉數十億美元級別的資產負債表包下龐大發電量，二線廠商在 PJM 等熱門電網拿不到足夠並網容量，面臨嚴峻的「電力排擠效應」。

區域內高耗能的傳統製造業（鋼鐵、重化工業）：

儘管增容部分的建設由 Google 出資，但高達 2,700 MW 的電力合約仍鎖定了電網中龐大的供給資源。隨著整體區域備載容量緊縮，PJM 容量拍賣（Capacity Auction）價格水漲船高，可能進一步推高當地傳統工廠的用電成本。

Our outside-the-box thinking

「核電增容」正在實質截胡 SMR（小型模組化反應爐）的黃金窗口期：

市場共識普遍認為 SMR 是解決 AI 供電的終極救星，但 SMR 面臨審批週期長、首台商業化成本過高（FOAK 風險）的挑戰。Constellation 此次透過 11 座現有反應爐增容 890 MW，直言這相當於「在更短時間內憑空建出好幾座 SMR」。這意味著在 2028～2035 年間，「增容 + 重啟舊機組」目前是科技業解決方案，甚至可能使部分商業化進度落後的 SMR 新創公司面臨資金被抽離的生存危機。

「自帶電力（BYOP）」表面上避免推高民生電價，實質上在將公共電網「階級私有化」：

儘管科技業標榜自費升級電廠不影響一般用電戶，但背後本質是頂級科技資本以 20 年長約包下全美最優質、最穩定的無碳基載發電能力。當優質核電被矽谷巨頭鎖定，公共電網與一般大眾將被迫分攤高波動、依賴天然氣或間歇性風光的發電風險。這在未來數年極可能引發嚴重的監管審查與社會反壟斷反彈（Regulatory & Social Backlash）。

科技巨頭正演變為「影子綜合能源公用事業（Shadow Utilities）」：

Google 這次不是被動向公用事業買電，而是扮演「電廠重大資本改造的出資方兼推手」。科技公司的資產負債表已直接穿透至發電基礎設施底層。未來衡

量一間 AI 超大規模雲端巨頭的核心壁壘，可能不再單看演算法參數或 GPU 庫存，而是「能源專案融資與物理鎖定基載電力的能力」。

極端能源溢價將反向倒逼「算力能效革命」與「端側/邊緣 AI」超預期爆發：

當鎖定幾座資料中心的電力需動輒砸下 40~50 億美元與 20 年承諾時，雲端暴力堆疊參數量（Brute-force Scaling）的邊際經濟效益正在逼近物理極限。這將逼迫產業加速尋找替代方案：更極致的稀疏化模型（MoE）、光學計算晶片，以及能將計算負載分散至終端設備的「邊緣/端側 AI（Edge AI）」，這反而可能成為下一輪顛覆雲端資料中心商業模式的非共識催化劑。

附註：什麼是 PJM 電網架構

PJM（PJM Interconnection） 是全美規模最大、運作機制最成熟的區域輸電組織（RTO, Regional Transmission Organization），負責調度與協調美國東部及中西部 13 個州以及華盛頓特區的高壓電網與電力批發市場。

其名稱源自最初結盟的三個州：Pennsylvania（賓夕法尼亞）、Jersey（紐澤西）、Maryland（馬里蘭）。如今，包含北維吉尼亞州（全球最密集的資料中心聚落 Data Center Alley）都在 PJM 的管轄範圍內。

一、 PJM 的組織角色與定位

PJM 本身不擁有電廠、不擁有電網，也不從事電力買賣。它是一個受美國聯邦能源監管委員會（FERC）監管的獨立非營利實體，核心角色有二：

電網可靠度調度員（Grid Reliability Operator）：控制超過 84,000 英里的高壓輸電線路，秒級平衡區域內 6,500 萬人的用電需求與發電供給。

批發電力市場管理者（Market Operator）：營運透明、競爭性的電力批發市場，媒合獨立發電廠（如核電、燃氣、風光發電商）與電力零售商/大用電戶。

二、 PJM 的三大市場架構

PJM 的電力市場運作主要依賴三大核心機制：

電能市場（Energy Market）：

分為日前市場（Day-Ahead）與即時市場（Real-Time）。

採用分區邊際定價（LMP, Locational Marginal Pricing）。LMP 不僅反映發電成本，還包含輸電線路壅塞成本（Congestion）與線路損耗（Losses）。若某區域（如維吉尼亞機房區）用電需求暴增且輸電線路壅塞，當地的 LMP 就會飆高。

容量市場（Capacity Market / RPM, Reliability Pricing Model）：

這是 PJM 最具特色的機制。為確保「未來 3 年」電網在尖峰負載時絕不斷電，PJM 每年舉行容量基礎拍賣（BRA, Base Residual Auction）。

發電廠競標承諾在未來特定年份提供足夠的備用容量，得標者即使不發電，也能按日領取「保證在場費」。近來因老舊電廠退役與 AI 負載激增，PJM 容量拍賣清算價格屢創新高。

輔助服務市場（Ancillary Services Market）：

採購調頻（Frequency Regulation）、備轉容量（Spinning Reserves）等服務，確保電網在發電機跳機或負載突增時，能在數秒至數分鐘內維持電網頻率（60Hz）穩定。

三、 為什麼 AI 巨頭與核電交易都繞不開 PJM？

當前科技巨頭（微軟、Google、Meta、亞馬遜）的核電採購與大型資料中心擴建，高度集中在 PJM 範圍內，主因在於其獨特的戰略位置與市場結構：

資料中心的心臟地帶：全美乃至全球伺服器密度最高的維吉尼亞州「Data Center Alley」以及鄰近的俄亥俄州、賓州，均位於 PJM 轄區內，AI 帶來的增量用電需求首當其衝撞擊 PJM 電網。

最豐富的非受監管核電資產：PJM 轄區內坐落著全美規模最大的民營核電機組群（主要由 Constellation、Vistra、PSEG 營運）。在解除管制的市場架構下，發電商能自由與科技巨頭簽署雙邊購電協議（PPA）或進行機組增容投資，無需經過各州公用事業委員會漫長且嚴苛的民生電價管制審查。

並網壅塞（Interconnection Queue）成為核心瓶頸：大量新能源專案在 PJM 等待並網審查，排隊時間動輒 5 年以上。科技巨頭為了避免機房「有晶片卻沒電開機」，只能轉向在 PJM 境內尋找「已並網的既有核電廠」進行功率擴增（Uprates）或退役機組重啟，藉此繞過漫長的電網排隊程序。

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